PYTHON FOR BEGINNERS · SESSION 1

Introduction to Python and Setup

Install Python, set up your editor, and run your first program.

What is Python?

Python is a **high-level, interpreted, general-purpose programming language** with **simple, English-like syntax**. You can read a Python program almost like a sentence, which is why it is **the most common first language**.

- **High-level:** you write human-friendly instructions and Python handles the machine details such as memory.
- **Interpreted:** code runs line by line, with no separate compile step. You write it, run it, and see the result straight away.
- **Dynamically typed:** you do not declare variable types. Python works out whether a value is a number, text or something else.
- **General-purpose:** one language, many jobs, instead of being limited to one kind of software.

What Python is used for

- Data analysis and visualisation
- Artificial intelligence and machine learning
- Web development with frameworks such as Django and Flask
- Automation and scripting of repetitive tasks
- Games
- Cybersecurity

Why it suits beginners

- **Short code:** a task that takes many lines in other languages often takes a few in Python.
- **A huge community,** so almost every question you will have has already been answered.
- **Thousands of libraries,** including Pandas (tables of data), NumPy (fast numbers) and Matplotlib (charts).

History





Python was created by **Guido van Rossum** in the Netherlands. He started working on it in **late 1989** and the **first release came in 1991**.

The name comes from the British comedy show **Monty Python's Flying Circus**, **not from the snake**. The snake logo came later.

YEAR	MILESTONE
1989	Guido van Rossum starts work on Python
1991	First public release
2000	Python 2.0 released
2008	Python 3.0 released
1 Jan 2020	Python 2 reaches end of life

> **Always install Python 3**
> Python 2 is no longer supported, so everyone uses Python 3 now. Always install the **latest stable Python 3 release**.

Installation

Windows

1. Go to python.org, open Downloads, and get the latest Python 3.
2. Run the installer and tick **Add python.exe to PATH**.
3. Click **Install Now**.
4. Open Command Prompt and run the command below to check it worked.

Most common beginner mistake

Forgetting to tick “Add python.exe to PATH” on the first installer screen. Without it, the terminal cannot find Python.

Command Prompt

```
python --version
```

OUTPUT

```
Python 3.14.0
```

Windows: install from the terminal

Windows 10 and 11 include **winget**, so you can install Python with one command and skip the website. Open **PowerShell** or **Command Prompt** and run:

PowerShell

```
winget install -e --id Python.Python.3.14
```

Close the terminal and **open a new one** so it picks up the new PATH. Then check the install worked:

PowerShell

```
python --version
pip --version
```

OUTPUT

```
Python 3.14.0
```

To see which versions are available, or to update later:

PowerShell

```
winget search Python.Python
winget upgrade --id Python.Python.3.14
```

Try py if python is not found

Windows also installs the `py` launcher. If `python` is not recognised, `py --version` often works.

macOS

Install from python.org, or use **Homebrew**. Then check the version with `python3`.

Terminal

```
brew install python
python3 --version
```

Linux

```
Terminal
sudo apt install python3 python3-pip
python3 --version
```

VS Code (editor)

1. Install VS Code from code.visualstudio.com.
2. Open the Extensions panel and install the **Python** extension by Microsoft.
3. Create a folder for your work and open it in VS Code.
4. Make a new file called `hello.py`.

> **No-install backup: Google Colab**
> Google Colab (colab.research.google.com) runs Python in any browser, with nothing to install. It is useful if your laptop is slow or locked down.

VS Code: install from the terminal

On Windows you can install VS Code with **winget** too. Open **PowerShell** and run:

```
PowerShell
winget install -e --id Microsoft.VisualStudioCode
```

Open a **new terminal**, then check the `code` command works and add the Python extension from the terminal:

```
PowerShell
code --version
code --install-extension ms-python.python
```

Make a practice folder, create `hello.py`, open the folder in VS Code, and run the file:

```
PowerShell
mkdir python-practice
cd python-practice
New-Item hello.py
code .
python hello.py
```

> **macOS and Linux**
> On macOS use `brew install --cask visual-studio-code`. On Ubuntu use `sudo snap install code --classic`. The `code` commands above work the same.

Common problems

PROBLEM	FIX
`python` is not recognized	PATH was not ticked during install. Reinstall and tick it, or add Python to PATH manually.
`python` opens the Microsoft Store	Turn off the Python entries in Settings > Apps > Advanced app settings > App execution aliases, or use `py` instead.
`code` is not recognized	Open a new terminal first. If it still fails, reinstall VS Code and keep **Add to PATH** ticked.
Command not found on Mac or Linux	Use `python3` instead of `python`.

Ways to run Python

Interactive mode (REPL)

Type `python` in the terminal. You get a `>>>` prompt where **every line runs the moment you press Enter**. It is perfect for trying small ideas. Leave with `exit()`.

REPL

```
>>> 2 + 3
5
>>> exit()
```

Script mode

Write your code in a file such as `hello.py`, then **run the whole file** from the terminal. Use this for anything you want to keep.

Terminal

```
python hello.py
```

VS Code

Open your file and click the Run button (▶) at the top right. The output appears in the terminal below.

Jupyter and Colab

Cell-based notebooks where you run code in small pieces and see results under each cell. We use these later in the Data Analyst track.

First programs

hello.py

```
print("Hello, World!")
print("My name is Aman")
print("Python", "is", "easy")
print("Python", "is", "easy", sep="-")
print("Line 1\nLine 2")
```

OUTPUT

```
Hello, World!
My name is Aman
Python is easy
Python-is-easy
Line 1
Line 2
```

Key points

- `print()` is a **built-in function**. It shows whatever you give it on the screen.
- **Text (a "string") goes inside quotes**. Single or double quotes both work.
- You can give `print()` several values. It puts a space between them by default, and `sep=` changes that.
- **Case matters**: `print` works, `Print` gives an error.
- `\n` starts a new line and `\t` inserts a tab.

Comments

Comments are notes for people. **Python ignores them**. Use `#` for one line, or triple quotes for several lines.

```
# This is a single-line comment
"""
This is a multi-line comment
"""
```

Reading errors

Everyone gets errors, and they are useful. Break the code on purpose, read the message, then fix it.

Broken on purpose

```
print("Hello)        # SyntaxError: missing quote
Print("Hello")       # NameError: wrong case
```

ERROR

```
  File "your code", line 1
    print("Hello)        # SyntaxError: missing quote
          ^
SyntaxError: unterminated string literal (detected at line 1)
```

Read the error message from the bottom. It tells you the **line number** and the **error type**. A `SyntaxError` means the code is written wrongly. A `NameError` means Python does not recognise a name.

Python as a calculator

```
print(10 + 5)
print(10 / 3)
print(10 // 3)
print(10 % 3)
print(2 ** 3)
```

OUTPUT

```
15
3.3333333333333335
3
1
8
```

OPERATOR	MEANING	EXAMPLE	RESULT
`+`	Add	`10 + 5`	15
`/`	Divide, with a decimal result	`10 / 3`	3.3333333333333335
`//`	Divide, keep the whole part only	`10 // 3`	3
`%`	Remainder after dividing	`10 % 3`	1
`**`	Power	`2 ** 3`	8

Practice

1. Print your name, college and branch on three lines.
2. Print a simple box or star pattern using `print()`.
3. Print the result of `(25 * 4) + 10`.
4. Fix the broken code: `prnt("Hi")`.

Your turn: write your answers here, then press Run

```
# 1. Your name, college and branch
print("Your name")

# 4. Fix this line
prnt("Hi")
```

Homework and wrap-up

1. Install Python and VS Code (or use Colab) and run `hello.py`.
2. Write a program that prints a short self-introduction.
3. Send a screenshot of the output to the class group.

Next class

Variables and data types.